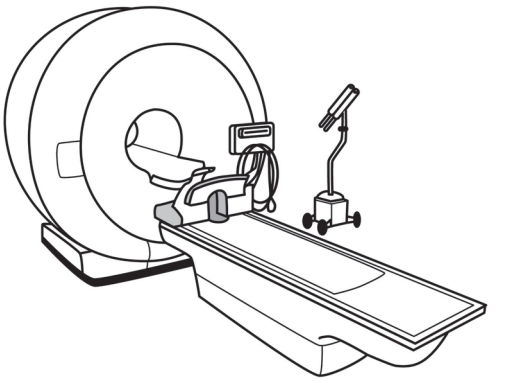
New imaging technology

Medical technology has improved greatly over time, and doctors now have many ways of looking inside patients without cutting them open.

X-rays don't pass through bone, so they can make an image of the bone like a shadow on a film or screen. X-rays can pass through soft tissue in varying amounts, making shadows that are weaker than bone images.

A **CAT** scan is a computer controlled series of X-rays that shows an area of the body in a series of 'slices'. This allows doctors to see tumours or injuries.



Magnetic Resonance Imaging machine

Ultrasound vibrations are used to make an image of an internal organ such as a heart, or of an unborn baby. **Magnetic Resonance Imaging** (MRI) uses large magnets and radio waves to make a three-dimensional image.

☞ **1** Draw a line matching each technique with its abbreviation.

MRI	Diffuse Optical Imaging – uses near infrared light
CAT	Event-Related Optical Signal – uses infrared light through optical fibres
MRE	Computed Axial Tomography – uses X-rays controlled by a computer
DOI	MagnetoEncephaloGraphy – uses magnetic field of the brain
EROS	Image-directed Colour Doppler Ultrasound – uses ultrasound vibrations
EEG	ElectroCardioGraphy – uses electric signals of the heartbeat
ECG	ElectroEncephaloGraphy – uses the brain's own electric signals
ICDU	Magnetic Resonance Elastography – uses huge magnets and radio waves
MEG	Magnetic Resonance Imaging – uses huge magnets and radio waves

2 Which technique could be used to show a brain in a series of slices?

Abbreviation: _____. Full name: _____.

3 To record the electrical pulse of the heart a doctor would use:

Abbreviation: _____. Full name: _____.

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